

Report
Of
Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003001343

Name of Student: SHIVANSHU MISHRA

Batch: 2025-29

2026-27

CANDIDATE'S DECLARATION

I, SHIVANSHU MISHRA do hereby declare that the internship report titled **“Python Development Intern”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name:- SHIVANSHU MISHRA

Roll No. :- 25SCS1003001343

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name:- SHIVANSHU MISHRA

Roll No. :- 25SCS1003001343

COGNIFYZ TECHNOLOGIES • INTERNSHIP PORTFOLIO

Python Development Internship Report

A Comprehensive Overview of 13 Technical Projects Across Core Scripting, Web Scraping, Data Analytics & OS Automation

INTERN NAME

Shivanshu Mishra

INTERN ID & INSTITUTION

CTI/A1/C403017 • IILM

INTERNSHIP TENURE

June 2, 2026 – July 1, 2026

Executive Overview

Demonstrating technical mastery, exception handling, and algorithmic problem-solving across 3 progressive tiers of software engineering.

| 13 Core Engineering Projects

13

Tasks Completed

Progressive Competency Curriculum

The internship program covered 3 structured competency tiers:

- **Level 01:** Core Python Syntax, Memory Index Traversal & Defensive Logic.
- **Level 02:** Algorithms, CLI State Machines, Cryptographic Password Evaluation & File Streams.
- **Level 03:** Web Mining (BeautifulSoup), Data Science (Pandas/Plotly) & OS System Automation.

Master Execution Matrix

Level / ID	Project Title	Key Computational Concepts	Domain
Level 01 (T1-T5)	Core Syntax & Logic	Negative Index Loops, Formula Mapping, Email Parsing	Core Python
Level 02 (T1-T2)	CLI Guessing Games	Random State Machines, Dynamic Interactive Loops	Algorithms
Level 02 (T3-T5)	Security & File Streams	Entropy Scoring, O(n) Fibonacci, Hash Map Word Count	Data & Files
Level 03 (T1-T3)	Scraping, Data & OS	DOM Web Parsing, Pandas/Plotly Charts, Shutil Sorter	Advanced Tech

Level 01: Foundations

Core Syntax & String Engines

Designed manual algorithms for string reversal and symmetrical palindrome scanning without relying on black-box slicing shortcuts.

Constructed a thermodynamic unit converter ($^{\circ}\text{C} \leftrightarrow ^{\circ}\text{F}$) and a zero-division defensive calculator to guard against runtime fatalities.



Level 01 Technical Mechanics



String Reversal

Iterative pointer accumulation loop.

```
for i in range(len(s)-1, -1, -1):  
    result += s[i]
```



Email Validator

Token split & domain suffix check.

```
portion = email.split("@")  
return "." in portion[1]
```



Defensive Math

Guards against division-by-zero.

```
if num2 == 0:  
    print("Cannot divide by zero")
```

| Level 02: Algorithms & Streams

Password Strength Evaluator

Multi-vector entropy scoring evaluating length (≥ 8), uppercase, lowercase, numeric digits, and special characters to classify passwords into Strong, Medium, or Weak.

File Stream Frequency Count

Constructed a persistent file parser reading external text files, normalizing case sensitivity, and generating alphabetical word counts using Python dictionaries.

Level 03: Data Analytics

Dynamic Visual Analytics

Built an interactive data visualization pipeline utilizing Pandas for CSV ingestion and Plotly Express for rendering dynamic graphical charts:

- Automatic schema header inspection
- Multi-chart support (Bar, Line, Scatter)
- Interactive browser presentation



Level 03: Web & OS Tasks

Automation & Web Mining

Engineered an automated web scraper using Requests and BeautifulSoup4 to parse HTML DOM trees and harvest structural content.

Built a system utility utilizing `os` and `shutil` to catalog disk files into dynamic extension categories (Images, Documents, Videos).



Quantitative Performance Impact



Transitioning manual text and file handling into automated Python pipelines produced massive efficiency gains across all performance metrics.

| Core Competencies Acquired

- ✓ **Core Python & Algorithmic Design:** Mastery of string mechanics, state loops, list comprehensions, and functional encapsulation.
- ✓ **Data Science & Analytics:** Hands-on experience with Pandas DataFrames and interactive Plotly visualization components.
- ✓ **Web Harvesting:** Practical DOM traversal and HTTP lifecycle management using BeautifulSoup4 and Requests.
- ✓ **OS System Automation:** Automated file-system directory sorting and file manipulation via native Python utilities.

Thank You!

Python Development Internship Presentation

Shivanshu Mishra • Intern ID: CTI/A1/C403017

Cognifyz Technologies • IILM University

Image Sources



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